A LOCALLY QUASI-FINITE MORPHISM OVER A QUASI-DM MORPHISM IS LOCALLY QUASI-FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter source sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Selection provenance: this card was opened under the file-based control contract (CONTROL.md), with the mechanical pre-card gate run on its statement and proof interval before the card was written.

Allowed standard prerequisites (editorial).

- The base change permanence of being locally quasi-finite. Invoked inside the counted proof (source lines 5229-5231). Makes the projection of the fibre product locally quasi-finite.
- The statement that the semi-diagonal of a quasi-DM morphism is locally quasi-finite. Invoked inside the counted proof (source lines 5232-5233). Makes the factorization arrow locally quasi-finite.
- The composition permanence of being locally quasi-finite. Invoked at the end of the counted proof (source lines 5234-5236). Combines the two factors into the assertion.

Source reference. Original lemma lemma-base-change-locally-quasi-finite: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5050–5050. Navigation pointer only: the counted text refers to the Stacks item lemma-base-change-locally-quasi-finite, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original lemma lemma-composition-locally-quasi-finite: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 5160–5160. Navigation pointer only: the counted text refers to the Stacks item lemma-composition-locally-quasi-finite, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Source reference. Original lemma lemma-semi-diagonal: sources/stacks-algebraic-stacks-morphisms/pinned-original-source.tex, source lines 845–845. Navigation pointer only: the counted text refers to the Stacks item lemma-semi-diagonal, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let $\mathcal{X} \to \mathcal{Y} \to \mathcal{Z}$ be morphisms of algebraic stacks. Assume that $\mathcal{X} \to \mathcal{Z}$ is locally quasi-finite and $\mathcal{Y} \to \mathcal{Z}$ is quasi-DM. Then $\mathcal{X} \to \mathcal{Y}$ is locally quasi-finite.*

Proof. Write $\mathcal{X} \to \mathcal{Y}$ as the composition

$$\mathcal{X} \longrightarrow \mathcal{X} \times_{\mathcal{Z}} \mathcal{Y} \longrightarrow \mathcal{Y}$$

The second arrow is locally quasi-finite as a base change of $\mathcal{X} \to \mathcal{Z}$, see Lemma lemma-base-change-locally-quasi-finite. The first arrow is locally quasi-finite by Lemma lemma-semi-diagonal as $\mathcal{Y} \to \mathcal{Z}$ is quasi-DM. Hence $\mathcal{X} \to \mathcal{Y}$ is locally quasi-finite by Lemma lemma-composition-locally-quasi-finite. □